Flows

Challenges of this module

* Not only costs on the arcs but also capacities
* applications: evacuation plans, scheduling, yield management, job assignment
* characterization of the optimal solution
* solution algorithm: improving a feasible solution
* minimum cost flow as a general framework
* can we generalize it to the case of multiple commodities?

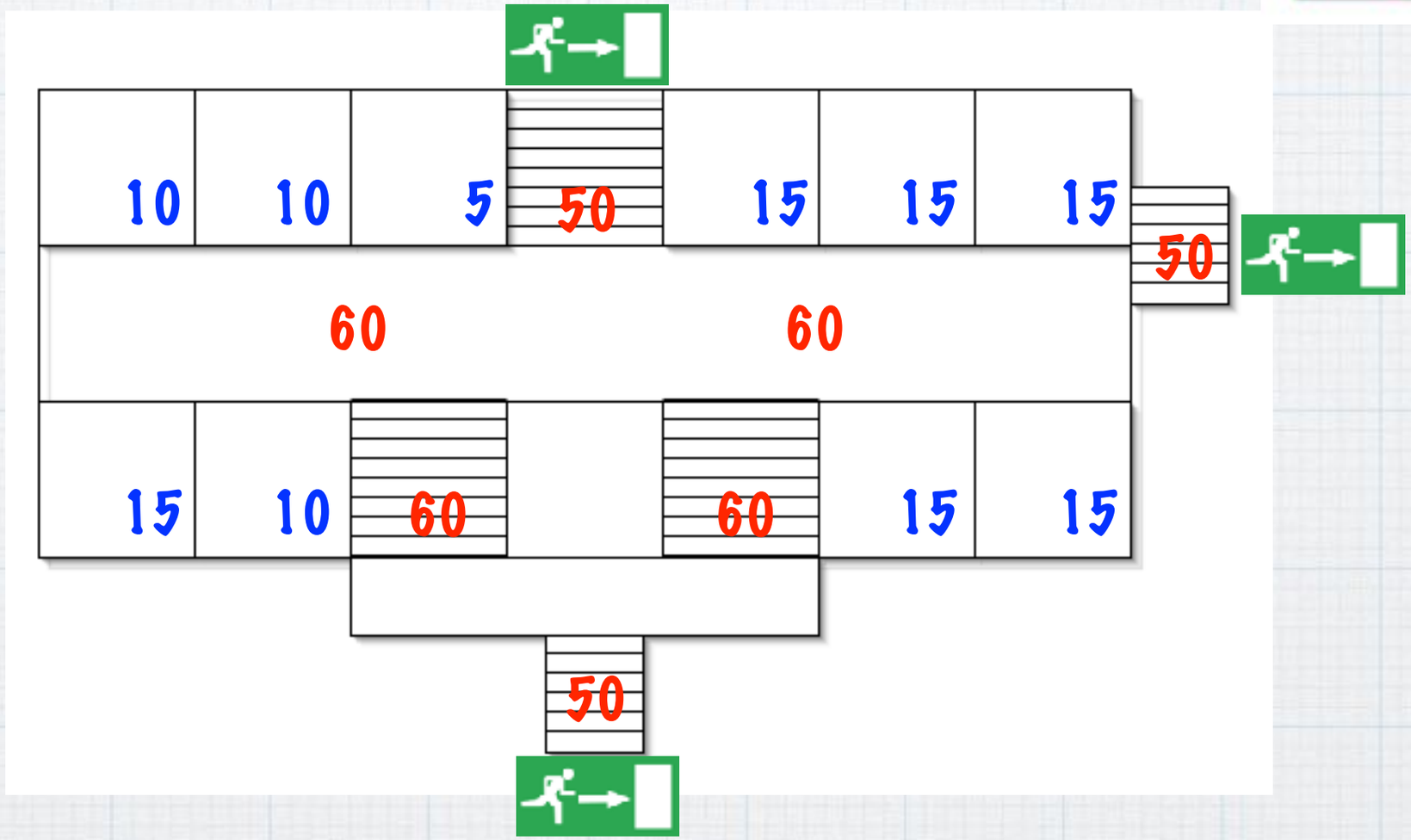
Link with previous knowledge

* search of a path in a graph
* either there is a path or there is a cut
* "dual" vision of the problem
* modeling with graphs (decisions on the arcs)
* detect negative cost cycles in a graph

Building evacuation plan

people to evacuate from the offices

corridors and stairways capacity



For each person determine the path to follow to reach the meeting point

Do not violate the capacity

Maximum flow problem

Given a directed graph $G = (N, A)$

and two nodes $s, t \in N$

(integer) capacities $u_{ij} \geq 0, \forall (i,j) \in A$

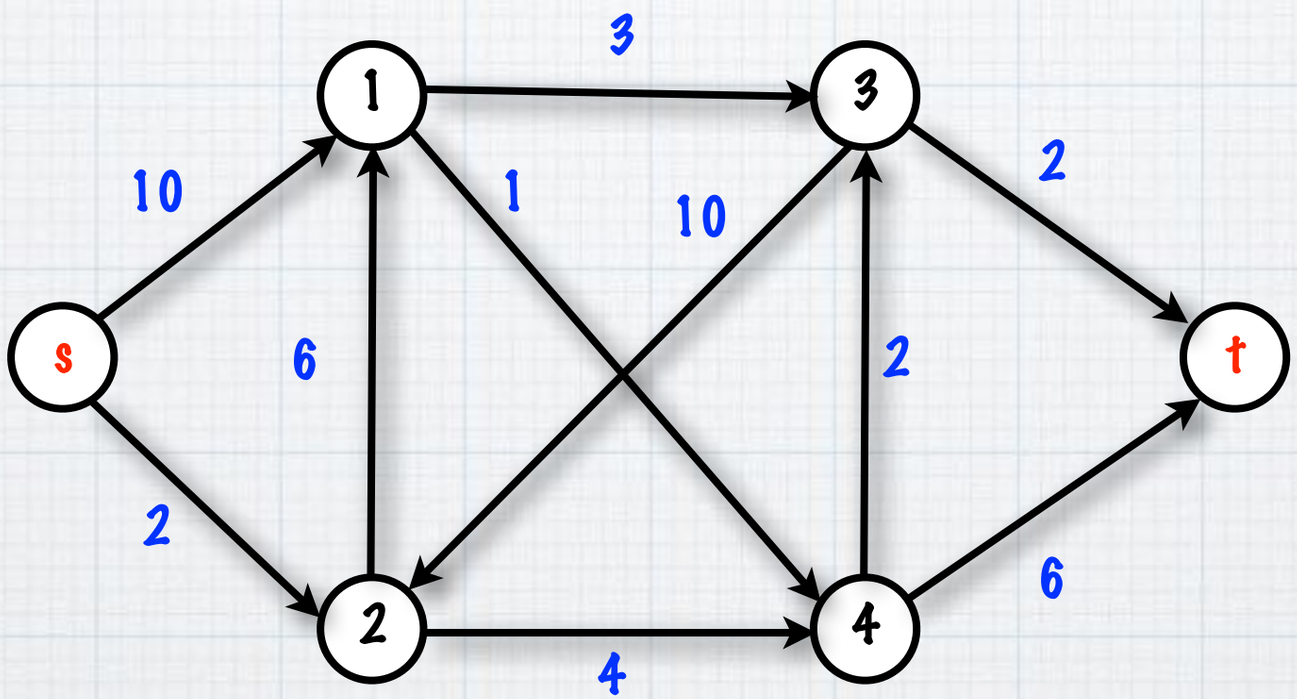
Send as much flow as possible from s to t

constraints

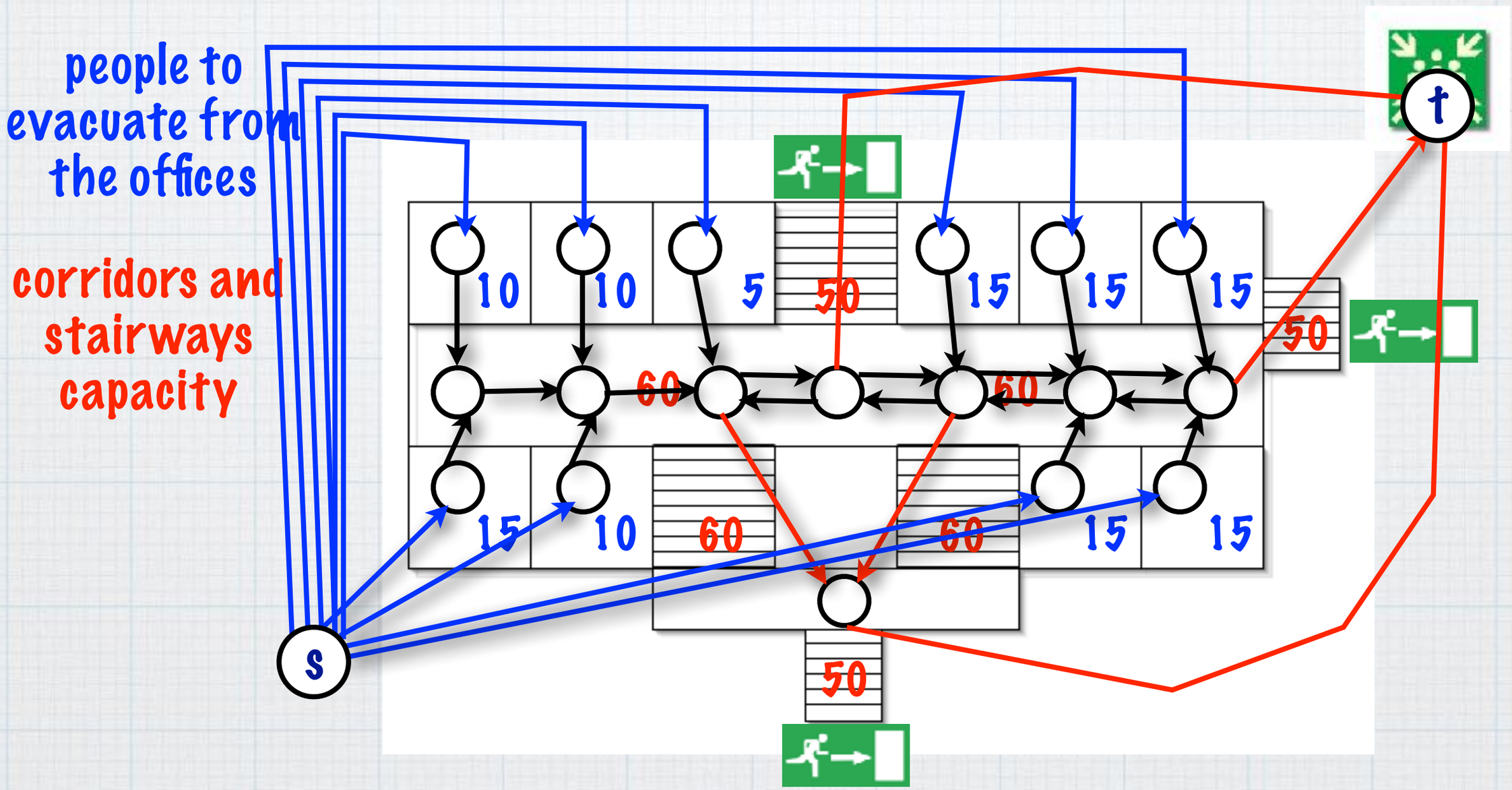
arc capacities

flow conservation at intermediate nodes

Example



Building evacuation plan



Scheduling problem

m identical machines

set of jobs $J = \{1, \ldots, n\}$

$$(p_j, r_j, d_j) \forall j \in J$$

processing time — release time — due date

jobs executed on a single machine

they can be interrupted and resumed without loss of time

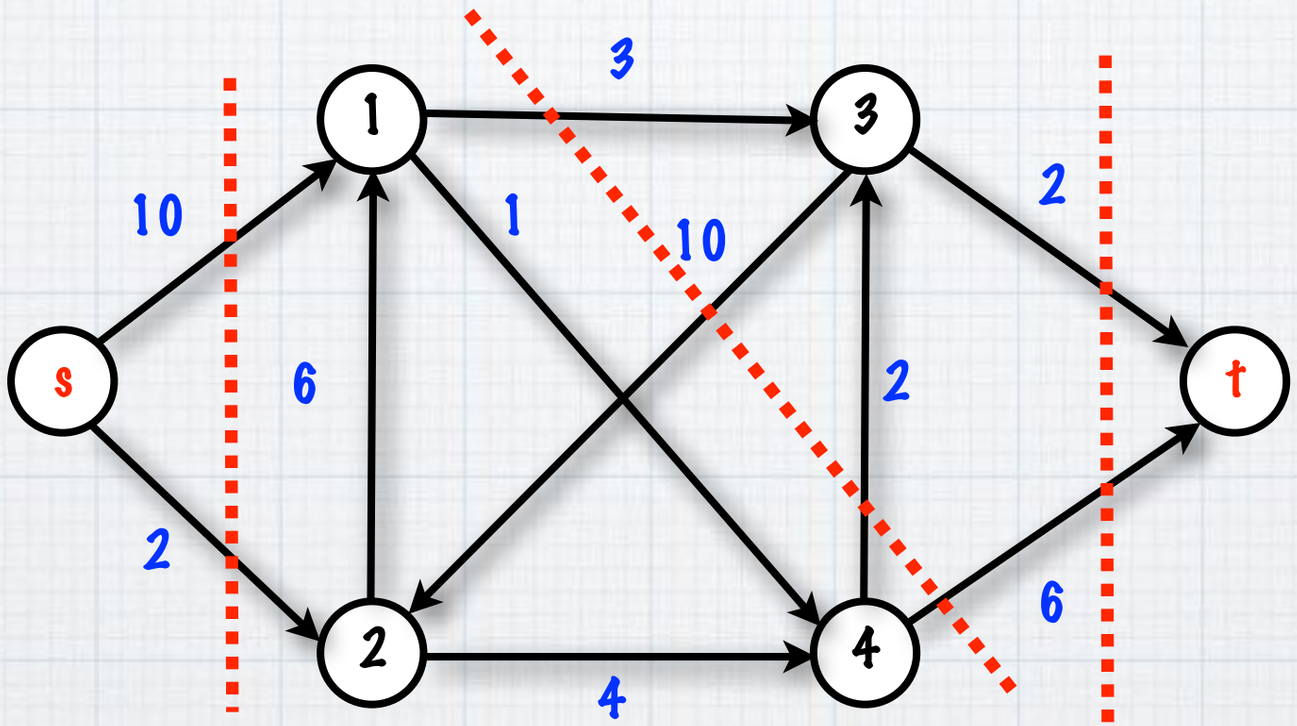
Determine an assignment job-machine

timing is satisfied

Alternative viewpoint

Interrupt the flow from s to t

destroy some arcs



cost proportional to the capacities

s-t cut

Given a directed graph $G = (N, A)$
and two nodes $s, t \in N$
(integer) capacities $u_{ij} \geq 0, \forall (i,j) \in A$

Cut (N_s, N_t)

$$N_s \cup N_t = N, N_s \cap N_t = \emptyset, s \in N_s, t \in N_t$$

Capacity of the cut

$$U(N_s, N_t) = \sum_{(i,j) \in A : i \in N_s, j \in N_t} u_{ij}$$

Flow traversing the cut

$$x(N_s, N_t) = \sum_{(i,j) \in A : i \in N_s, j \in N_t} x_{ij} - \sum_{(j,i) \in A : j \in N_t, i \in N_s} x_{ji}$$

$A^+(N_s, N_t)$ $\qquad$ $A^-(N_s, N_t)$

Properties

Given a feasible flow $x_{ij}, \forall (i,j) \in A$
for any cut (N_s, N_t)

the flow going from s to t equals the flow traversing the cut

$$v = x(N_s, N_t)$$

the flow traversing the cut is less than or equal to the capacity of the cut

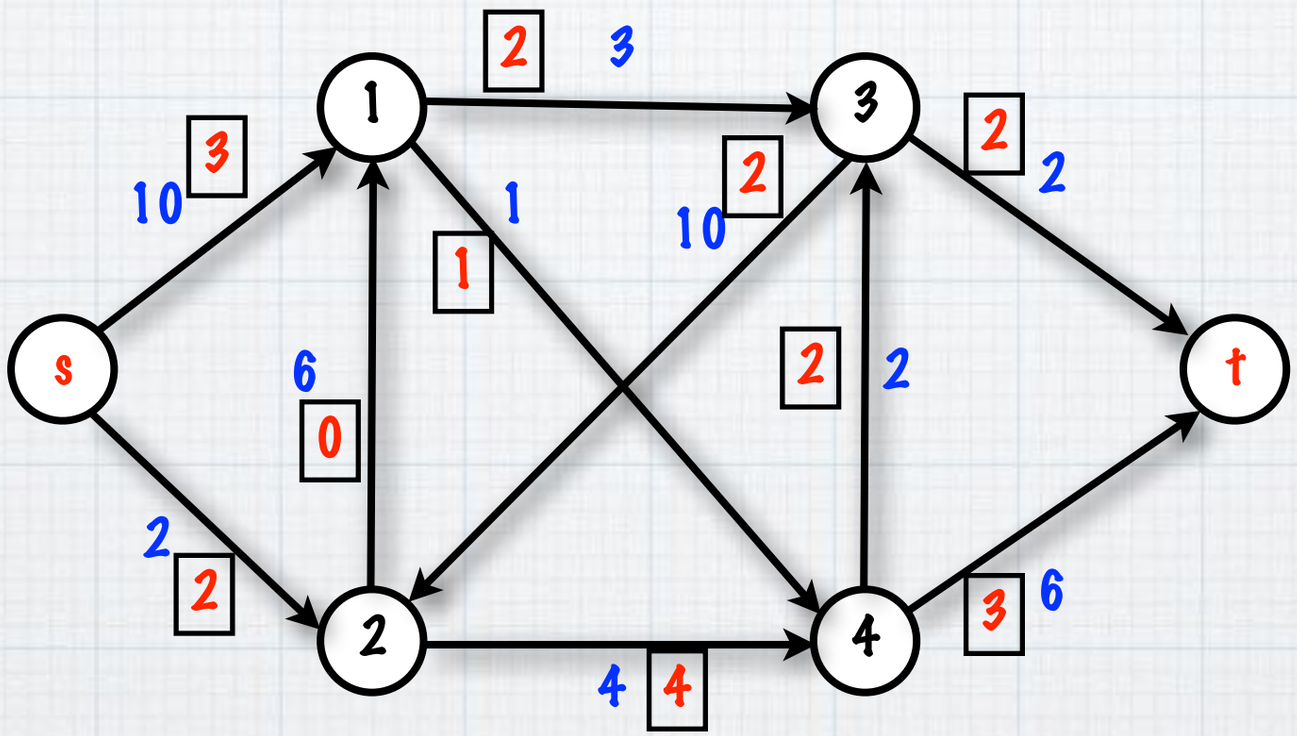
$$x(N_s, N_t) \leq U(N_s, N_t)$$

=

Can we prove that when x is optimum the equality holds?

Computation tool

Given a feasible flow



Can we improve the solution?

What are the possible changes?

Residual graph

Consider $G = (N, A) \quad s, t \in N$

(integer) **capacities** $u_{ij} \geq 0, \forall (i,j) \in A$

feasible flow $x_{ij}, \forall (i,j) \in A$

$$G_R(x) = (N, A(x) = A^+(x) \cup A^-(x))$$

$$A^+(x) = \{(i,j) \in A : x_{ij} < u_{ij}\}$$

the flow can be increased by at least one unit

$$A^-(x) = \{(j,i) : (i,j) \in A : x_{ij} > 0\}$$

the flow can be decreased by at least one unit

Combine the changes

Path from s to t in $G_R(x)$

$\updownarrow$

flow augmentation

Can we prove that if there are no paths the flow is maximum?

Augmenting_paths (G, s, t, x)

```
foreach (i,j) ∈ A do x[i,j]←0
repeat
    G_R ←Residual graph(x);
    Graph_search(G_R,s,P);
    if P[t]≠0 then  Augment_flow(P,x);
until P[t]=0
```

```
Augment_flow(P,x)
ϑ←Residual_capacity(P,x)
foreach (i,j) ∈ P do if (i,j) ∈ A⁺ then x[i,j] ← x[i,j]+ϑ
                     else x[j,i] ← x[j,i]−ϑ
```

$$\text{Residual_capacity}(P,x)=\min\{u_{ij}-x[i,j]:\ (i,j)\in A^{+}\cap P$$
$$x[j,i]:\ (i,j)\in A^{-}\cap P\}$$

Very important property

Given a directed graph $G = (N, A)$

and two nodes $s, t \in N$

either there is a path p_{st}

or there is a cut $(N_s, N_t), s \in N_s, t \in N_t$

such that each arc of the cutset is $(i, j) : i \in N_t, j \in N_s$

Complexity issues

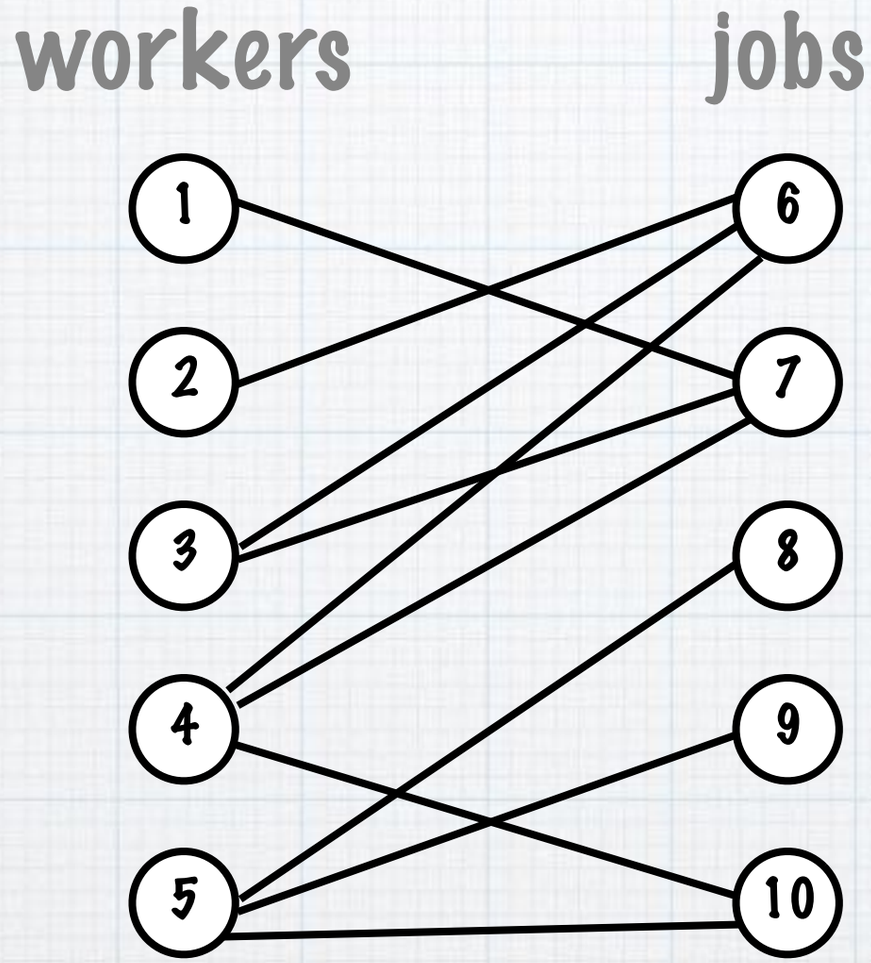
How to select the augmenting path

Edmonds and Karp

Select the augmenting path with the minimum number of arcs

$O(m^2n)$

Assignment problem



Compatibility

Assign jobs to workers (at most one worker per job)

Maximize the number of pairs

Maximum cardinality assignment

Given a bipartite graph $G = (S, T, A)$

$$A \subseteq S \times T$$

Matching $M \subseteq A$

no two arcs are incident in the same node

The problem can be easily reduced to a Maximum Flow

unit capacities

A waste disposal problem

10

pile of waste

cost, capacity

landfill

burner

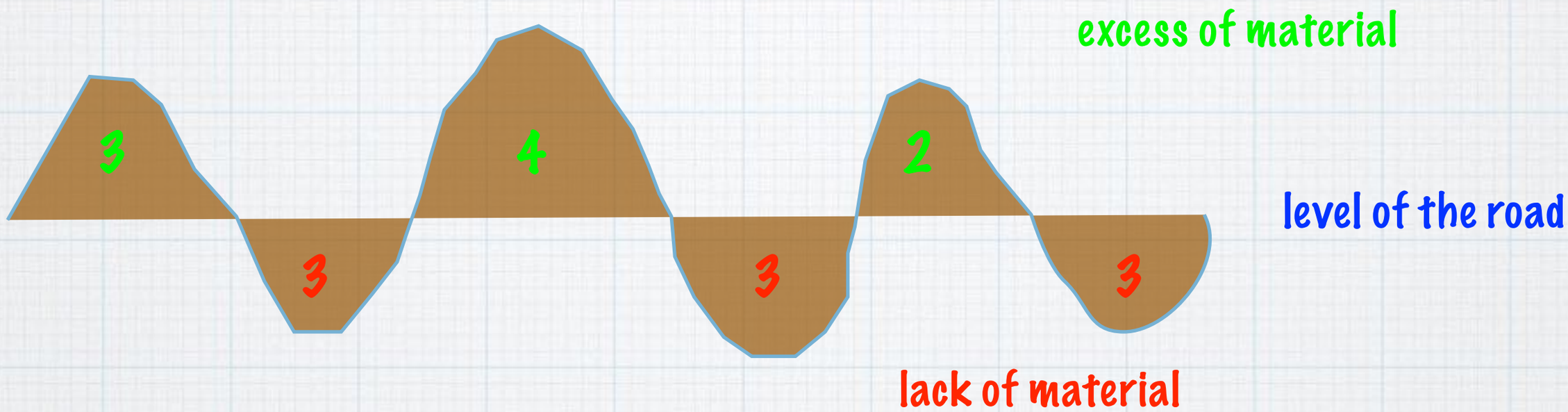
55

pile of waste

processing center

recycling center

Road construction



Cost of moving material from one site to an adjacent one = 1 per truck

Distribute the material at minimum cost

Pedemontana: estimated earth movement 35×10^6 m^3

Minimum cost flow problem

Given a graph $G = (N, A)$

node balances $b_i, \forall i \in N$

$b_i < 0$ offering node

$b_i > 0$ demanding node

$b_i = 0$ transfer node

w.l.o.g. let us assume that $\sum_{i \in N} b_i = 0$

costs $c_{ij}, \forall (i,j) \in A$

capacities $u_{ij} \geq 0, \forall (i,j) \in A$

send the required flow (flow balance constraints)

capacity constraints

minimize the cost of the flow

Generalization

Max Flow and **Shortest Path Tree** problem can be seen as **special cases** of the **Minimum Cost Flow Problem**

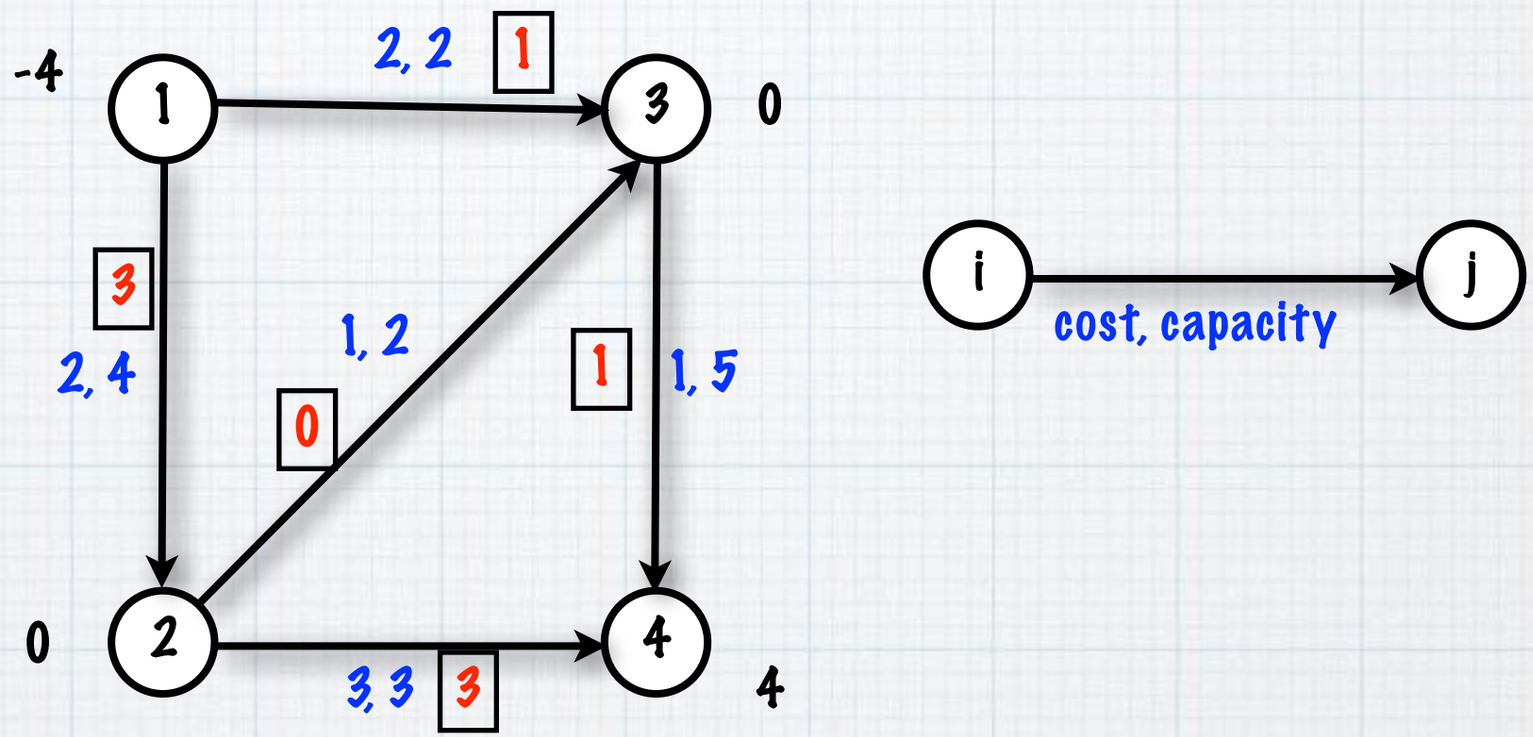
$$\min \sum_{(i,j)\in A} c_{ij}x_{ij}$$

$$\sum_{(j,i)\in BS(i)} x_{ji} - \sum_{(i,j)\in FS(i)} x_{ij} = b_i \quad \forall i \in N$$

$$0 \leq x_{ij} \leq u_{ij} \quad \forall (i,j) \in A$$

Changing a feasible solution

Given a feasible flow



Can we change the solution?

What changes preserve the feasibility?

What is the effect on the objective function of sending one unit of flow along a cycle?

Residual graph

Consider $G = (N, A)$

(integer) **capacities** $u_{ij} \geq 0, \forall (i,j) \in A$

costs $c_{ij}, \forall (i,j) \in A$

feasible flow $x_{ij}, \forall (i,j) \in A$

$$G_R(x) = (N, A(x) = A^+(x) \cup A^-(x))$$

$$A^+(x) = \{(i,j) \in A : x_{ij} < u_{ij}\}$$

the flow can be increased by at least one unit

cost: $c_{ij}, \forall (i,j) \in A^+(x)$

$$A^-(x) = \{(j,i) : (i,j) \in A : x_{ij} > 0\}$$

the flow can be decreased by at least one unit

cost: $-c_{ij}, \forall (j,i) \in A^-(x)$

Characterizing the optimal solution

If in the residual graph there is a **negative cost cycle**

the flow can be improved

We can prove that

If there are no negative cost cycles the flow is optimal

Negative cycle elimination

```
x←Feasible_flow;
G_R ←Residual graph(x);
while  Negative_cost_cycle(C, G_R)
    ϑ←Residual_capacity(C, x);
    Update_flow(ϑ, C, x);
    Update(G_R)
```

```
Update_flow(C, ϑ, x)
foreach (i,j) ∈ C do if (i,j) ∈ A⁺ then x[i,j] ← x[i,j]+ϑ
                     else  x[j,i] ← x[j,i]-ϑ
```

Detecting negative cost cycles

From the complexity analysis we know that a node cannot be inserted into Q more than n-1 times

If a node is inserted n times

negative cost cycle

Starting feasible flow

How can we determine a feasible flow (if any)?

Blended activity

Sections 7.3 and 7.4

Complexity issues

initialization → max flow

at each iteration: SPT_L_queue $O(mn)$

update the flow

update the graph $O(n)$

How many iterations at worst?

Selecting the cycle according to a particular rule

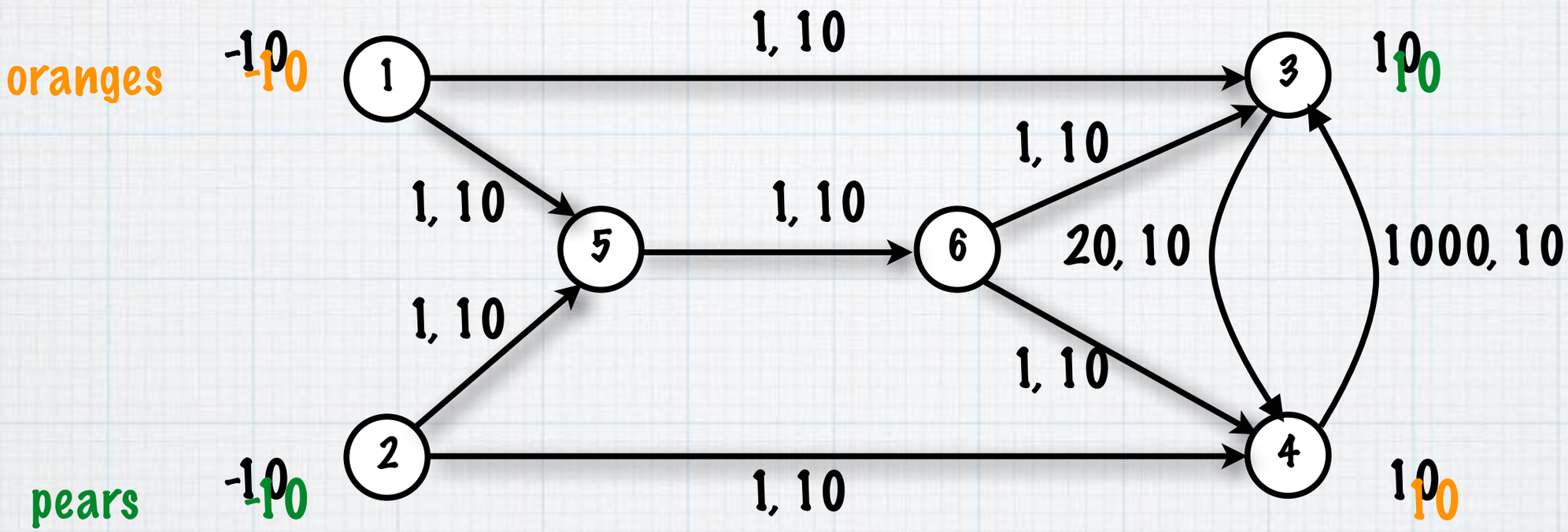
$O(m^2 n \log n)$ iterations

ILO

* Max flow-min cut property
* Apply EK augmenting path algorithm
* Apply the negative cycle elimination algorithm
* Determine a starting feasible flow
* Model with minimum cost flow problem

Chapter 2, section 6 and 7

A distribution problem



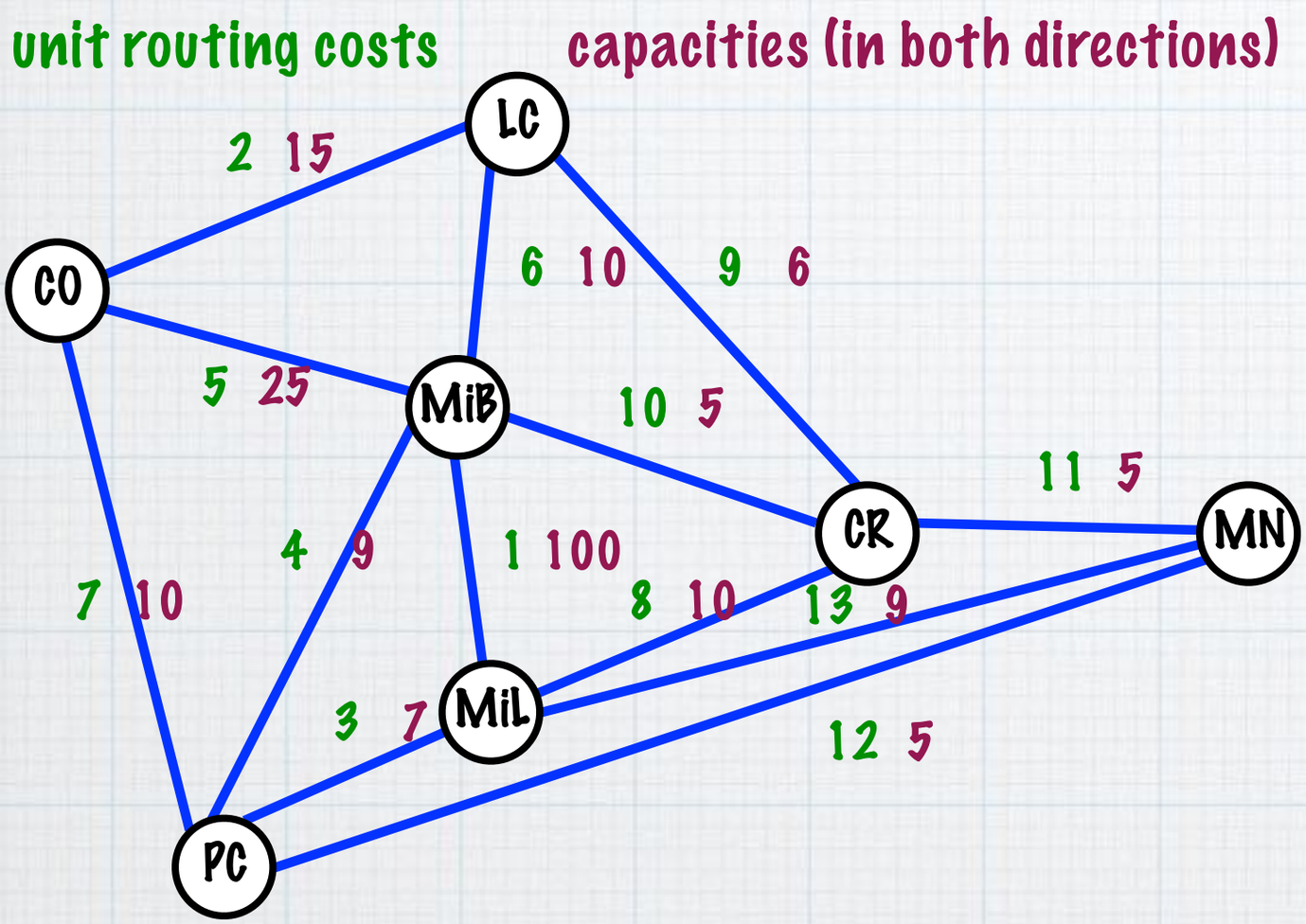
Multicommodity flow

A set of flow variables for each commodity

Flows of different commodities concur in the use of common resources (arc capacities)

A multi terminal routing problem

We want to route traffic the intranet network of Politecnico



Origin/Destination matrix

o/d	MiL	MiB	CO	LC	CR	PC	MN
MiL	-	80	30	0	15	2	10
MiB	70	-	0	20	0	5	5
CO	20	5	-	3	1	0	0
LC	1	3	1	-	2	0	1
CR	10	2	3	0	-	4	1
PC	2	5	0	0	3	-	0
MN	10	15	0	0	2	1	-

Problem:send the required flows from origins to destinations,
satisfy the capacities,
minimize the overall cost